

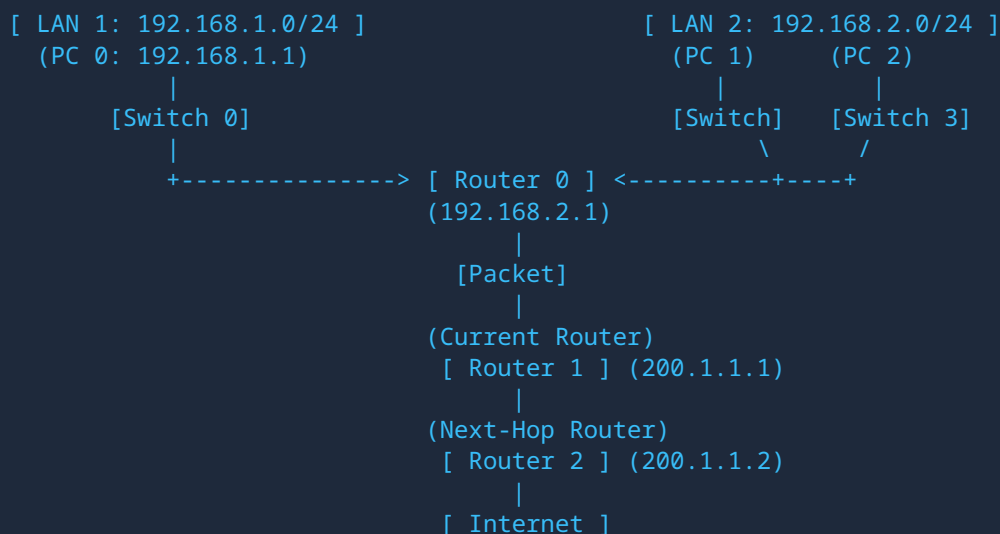
Network Infrastructure & Security Devices

1. Router

What is a Router?

A router connects different networks.

Router Network Diagram



What Does a Router Do?

- Connects a Local Area Network (LAN) to the Internet (LAN → Router → Internet).
- Routes packets based on IP addresses.
- Performs Network Address Translation or NAT (Private → Public IP).
- Maintains a routing table.

SOC Perspective & Logs:

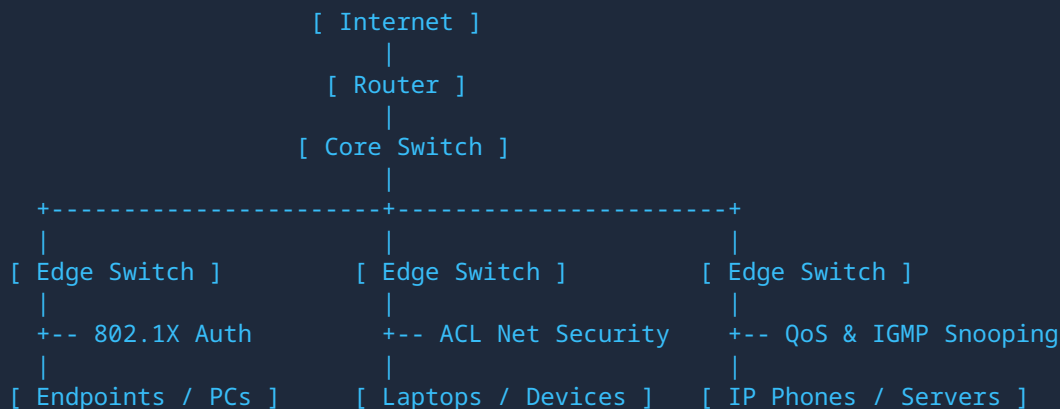
- **SOC Logs from Router:** Tracks Source IP, Destination IP, and Traffic flow.
- **SOC Indicators:** Traffic traveling to a malicious IP, unusual outbound network connections, or data exfiltration attempts.

2. Switch

What is a Switch?

A switch connects devices inside a single local network (LAN) (PC → Switch → Server).

Network Hierarchy & Switch Architecture Diagram



What Does a Switch Do?

- Uses hardware MAC addresses to route local traffic.
- Sends data exclusively to the correct destination device rather than broadcasting it.
- Implements security features like 802.1X Authentication, Access Control List (ACL) Network Security, and IGMP Snooping.

SOC Perspective & Threats:

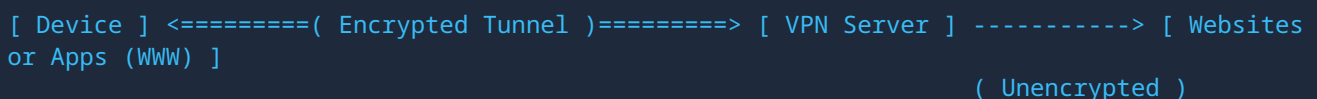
- **SOC Perspective:** Used to detect internal lateral movement and identify MAC address anomalies.
- **Threats:** ARP spoofing and MAC flooding.
- **SOC Example:** If the same MAC address is seen simultaneously on multiple switch ports, it indicates a possible Man-In-The-Middle (MITM) or ARP spoofing attack.

3. VPN (Virtual Private Network)

What is a VPN?

A VPN creates a secure, encrypted tunnel between a user and a network (User → Encrypted Tunnel → VPN Server → Internet / Internal Network).

VPN Tunneling Mechanism Diagram



What Does a VPN Do?

- Encrypts network traffic to ensure Confidentiality.
- Hides the user's real IP address to provide Anonymity.
- Allows secure remote access for corporate use cases.

Types of VPN

- **Remote Access VPN:** Connects an employee safely to a company network (e.g., Work from home setups).
- **SSL VPN:** Uses HTTPS (port 443) and is very common in corporate environments.
- **Site-to-Site VPN:** Connects entire offices together (Office A → Office B), primarily used across enterprises.

SOC Perspective & Indicators:

- **Normal Behavior:** User IP → VPN IP → Internal systems.
- **Suspicious Indicators:** Login attempts from an unusual country; multiple concurrent logins with the same credentials from different geographic locations; a VPN connection followed immediately by suspicious internal activity; unusually high data transfers.
- **Common VPN Abuse:** Attackers use VPNs maliciously to hide identity, bypass geo-restrictions, or access an internal network.
- **SOC Example (Impossible Travel):** User: admin | Login IP: Germany | VPN IP: India | Time gap: 2 minutes → **Compromised account.**

4. Firewall

What is a Firewall?

A firewall controls incoming and outgoing network traffic based on configured security rules. Its primary action is to Allow or Deny traffic.

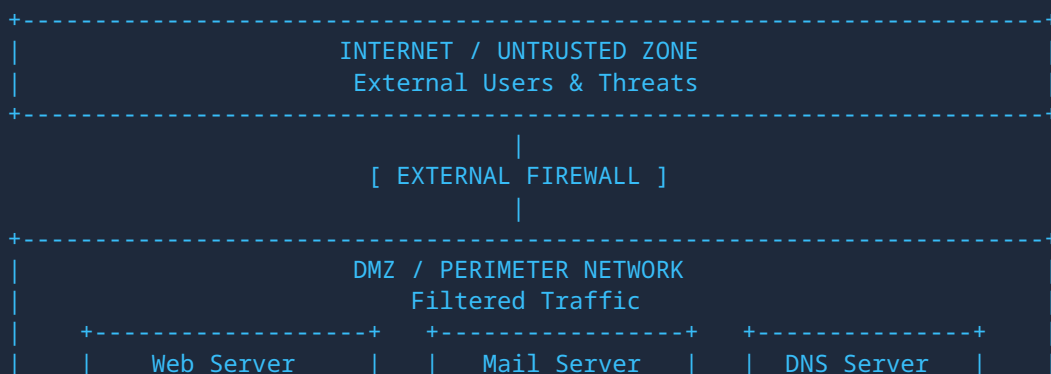
Firewall Actions Diagram

```
Incoming Packet ----> [ Firewall ] ----> Action 'Accept' ----> Packet permitted / passes through
Incoming Packet ----> [ Firewall ] ----> Action 'Deny' ----> Packet is silently dropped
Incoming Packet ----> [ Firewall ] ----> Action 'Reject' ----> 'Access denied' (ICMP prohibited / TCP RST sent back)
```

Types of Firewalls

Network Firewall | Host Firewall | Stateful Firewall

DMZ Network Segmentation Diagram

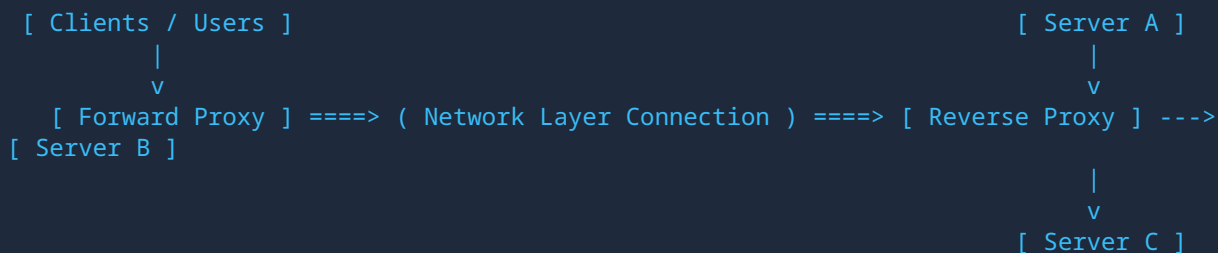


6. Proxy Servers

What is a Proxy Server?

A proxy acts as a middleman or intermediary gateway between an internal user and the internet (User → Proxy → Internet).

Forward vs. Reverse Proxies Diagram



- **Forward Proxy:** Sits in front of client devices to manage outbound requests, providing client anonymity and content/URL filtering. It serves the clients (users).
- **Reverse Proxy:** Sits directly in front of backend web servers to handle incoming customer requests, managing incoming load balancing, data caching, and infrastructure security. It serves the origin servers.

Proxy vs. VPN ??

- **VPNs:** Provide robust encryption (e.g., AES-256) and encrypt all internet traffic at the operating system level, making data completely unreadable to ISPs and hackers. They are ideal for total privacy and security.
- **Proxies:** Typically provide no encryption and function only at the application level (e.g., browser-only). They are faster and better suited for basic IP masking or bypassing simple geo-restrictions.

SOC Perspective & Logs:

- **SOC Logs:** Captures exact URLs accessed, specific user activity, and file upload/download behaviors.
- **SOC Indicators:** Access to known malicious domains, data exfiltration signatures, and unusual employee browsing habits.

Summary: Combining Everything Together

What the SOC Sees Across Devices

- **Router:** IP traffic
- **Switch:** MAC layer activity
- **Firewall:** Allowed vs. blocked traffic
- **IDS/IPS:** Real-time attack alerts
- **Proxy:** Detailed user web behavior

Real SOC Incident Scenario

Event: A user downloads an unknown file from an unclassified domain.

Device Logs:

- *Firewall:* Allowed
- *Proxy:* Flags a highly suspicious URL destination
- *IDS:* Detects known malware signatures

Conclusion: Active **Malware infection attempt**.

Analyst Critical Thinking

When a security alert triggers, determine its scope by asking **where did this happen?**

- **Internal Layer:** Look at the **Switch / MAC activity**.
- **Network Layer:** Look at the **Router / IP traffic**.
- **Security Perimeter:** Look at the **Firewall / IDS alerts**.
- **User Layer:** Look at the **Proxy / User web logs**.